In [1]:
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv(r"C:\Users\praka\Dataset .csv")
```

In [2]:
```
df.head()
```

Out[2]:

	Restaurant ID	Restaurant Name	Country Code	City	Address	Locality	Locality Verbose
0	6317637	Le Petit Souffle	162	Makati City	Third Floor, Century City Mall, Kalayaan Avenu...	Century City Mall, Poblacion, Makati City	Century City Mall, Poblacion, Makati City, Mak...
1	6304287	Izakaya Kikufuji	162	Makati City	Little Tokyo, 2277 Chino Roces Avenue, Legaspi...	Little Tokyo, Legaspi Village, Makati City	Little Tokyo, Legaspi Village, Makati City, Ma...
2	6300002	Heat - Edsa Shangri-La	162	Mandaluyong City	Edsa Shangri-La, 1 Garden Way, Ortigas, Mandal...	Edsa Shangri-La, Ortigas, Mandaluyong City	Edsa Shangri-La, Ortigas, Mandaluyong City, Ma...
3	6318506	Ooma	162	Mandaluyong City	Third Floor, Mega Fashion Hall, SM Megamall, O...	SM Megamall, Ortigas, Mandaluyong City	SM Megamall, Ortigas, Mandaluyong City, Mandal...
4	6314302	Sambo Kojin	162	Mandaluyong City	Third Floor, Mega Atrium, SM Megamall, Ortigas...	SM Megamall, Ortigas, Mandaluyong City	SM Megamall, Ortigas, Mandaluyong City, Mandal...

5 rows × 21 columns

In [3]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9551 entries, 0 to 9550
Data columns (total 21 columns):
 #   Column                Non-Null Count  Dtype
---  ------                --------------  -----
 0   Restaurant ID         9551 non-null   int64
 1   Restaurant Name       9551 non-null   object
 2   Country Code          9551 non-null   int64
 3   City                  9551 non-null   object
 4   Address               9551 non-null   object
 5   Locality              9551 non-null   object
 6   Locality Verbose      9551 non-null   object
 7   Longitude             9551 non-null   float64
 8   Latitude              9551 non-null   float64
 9   Cuisines              9542 non-null   object
 10  Average Cost for two  9551 non-null   int64
 11  Currency              9551 non-null   object
 12  Has Table booking     9551 non-null   object
 13  Has Online delivery   9551 non-null   object
 14  Is delivering now     9551 non-null   object
 15  Switch to order menu  9551 non-null   object
 16  Price range           9551 non-null   int64
 17  Aggregate rating      9551 non-null   float64
 18  Rating color          9551 non-null   object
 19  Rating text           9551 non-null   object
 20  Votes                 9551 non-null   int64
dtypes: float64(3), int64(5), object(13)
memory usage: 1.5+ MB
```

In [4]:
```
df.tail()
```

Out[4]:

	Restaurant ID	Restaurant Name	Country Code	City	Address	Locality
9546	5915730	Naml\ Gurme	208	××stanbul	Kemanke×× Karamustafa Pa××a Mahallesi, R\ht\m ...	Karak×_y
9547	5908749	Ceviz A××ac\	208	××stanbul	Ko××uyolu Mahallesi, Muhittin ××st×_nda×× Cadd...	Ko××uyolu
9548	5915807	Huqqa	208	××stanbul	Kuru×_e××me Mahallesi, Muallim Naci Caddesi, N...	Kuru×_e××me
9549	5916112	A×××k Kahve	208	××stanbul	Kuru×_e××me Mahallesi, Muallim Naci Caddesi, N...	Kuru×_e××me
9550	5927402	Walter's Coffee Roastery	208	××stanbul	Cafea××a Mahallesi, Bademalt\ Sokak, No 21/B, ...	Moda

5 rows × 21 columns

In [5]:
```
df.describe()
```

Out[5]:

	Restaurant ID	Country Code	Longitude	Latitude	Average Cost for two	Price range	9
count	9.551000e+03	9551.000000	9551.000000	9551.000000	9551.000000	9551.000000	
mean	9.051128e+06	18.365616	64.126574	25.854381	1199.210763	1.804837	
std	8.791521e+06	56.750546	41.467058	11.007935	16121.183073	0.905609	
min	5.300000e+01	1.000000	-157.948486	-41.330428	0.000000	1.000000	
25%	3.019625e+05	1.000000	77.081343	28.478713	250.000000	1.000000	
50%	6.004089e+06	1.000000	77.191964	28.570469	400.000000	2.000000	
75%	1.835229e+07	1.000000	77.282006	28.642758	700.000000	2.000000	
max	1.850065e+07	216.000000	174.832089	55.976980	800000.000000	4.000000	

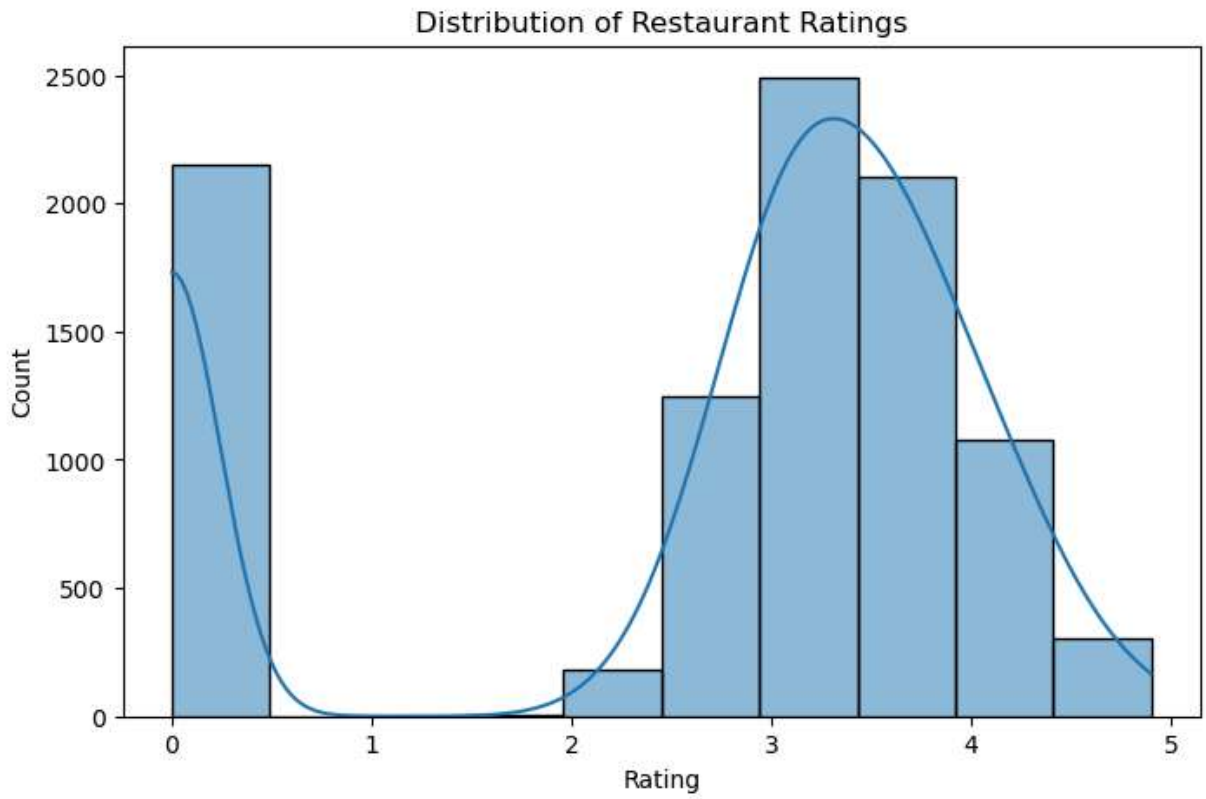
In [6]:
```
df.shape
```

Out[6]:
```
(9551, 21)
```

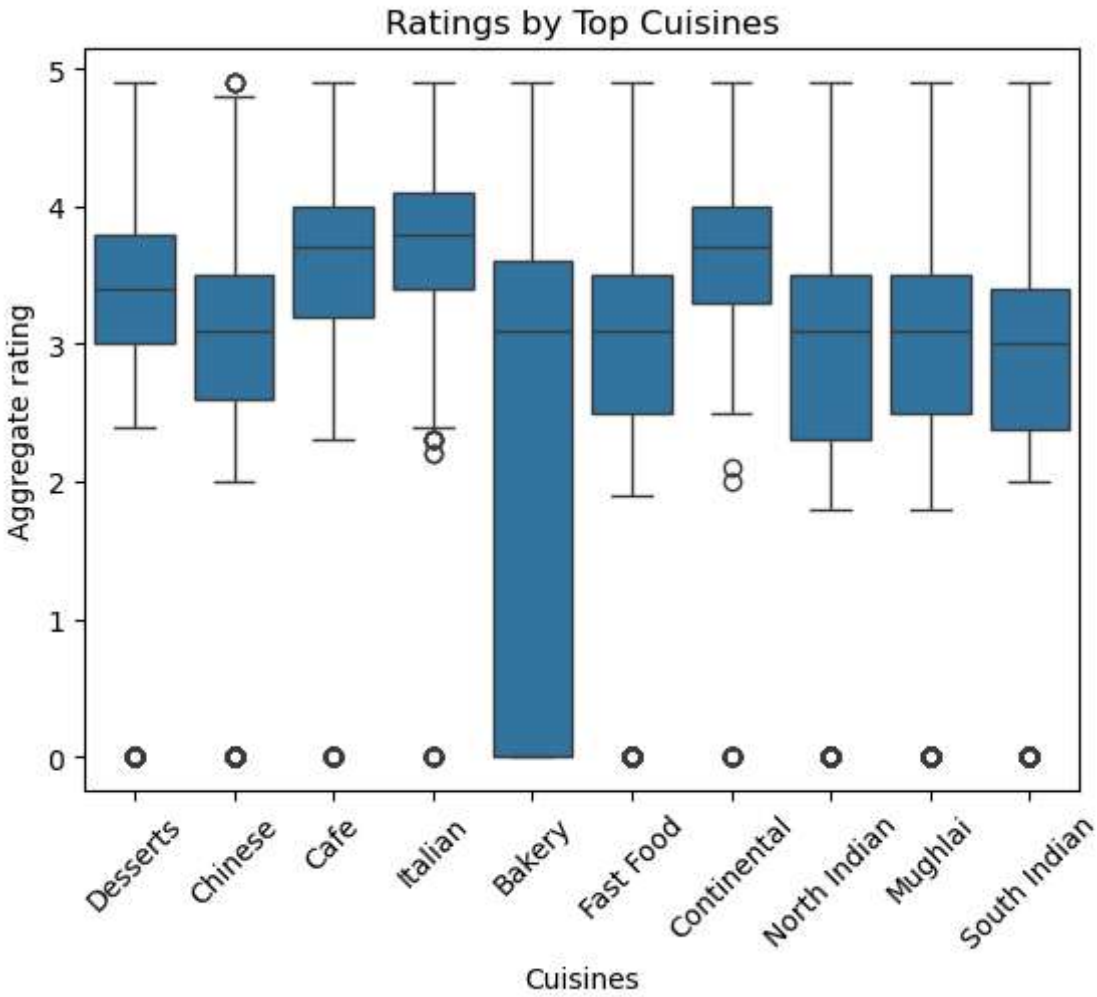
In [8]:
```
df.isnull().sum()
```

Out[8]:
```
Restaurant ID           0
Restaurant Name         0
Country Code            0
City                    0
Address                 0
Locality                0
Locality Verbose        0
Longitude               0
Latitude                0
Cuisines                9
Average Cost for two    0
Currency                0
Has Table booking       0
Has Online delivery     0
Is delivering now       0
Switch to order menu    0
Price range             0
Aggregate rating        0
Rating color            0
Rating text             0
Votes                   0
dtype: int64
```

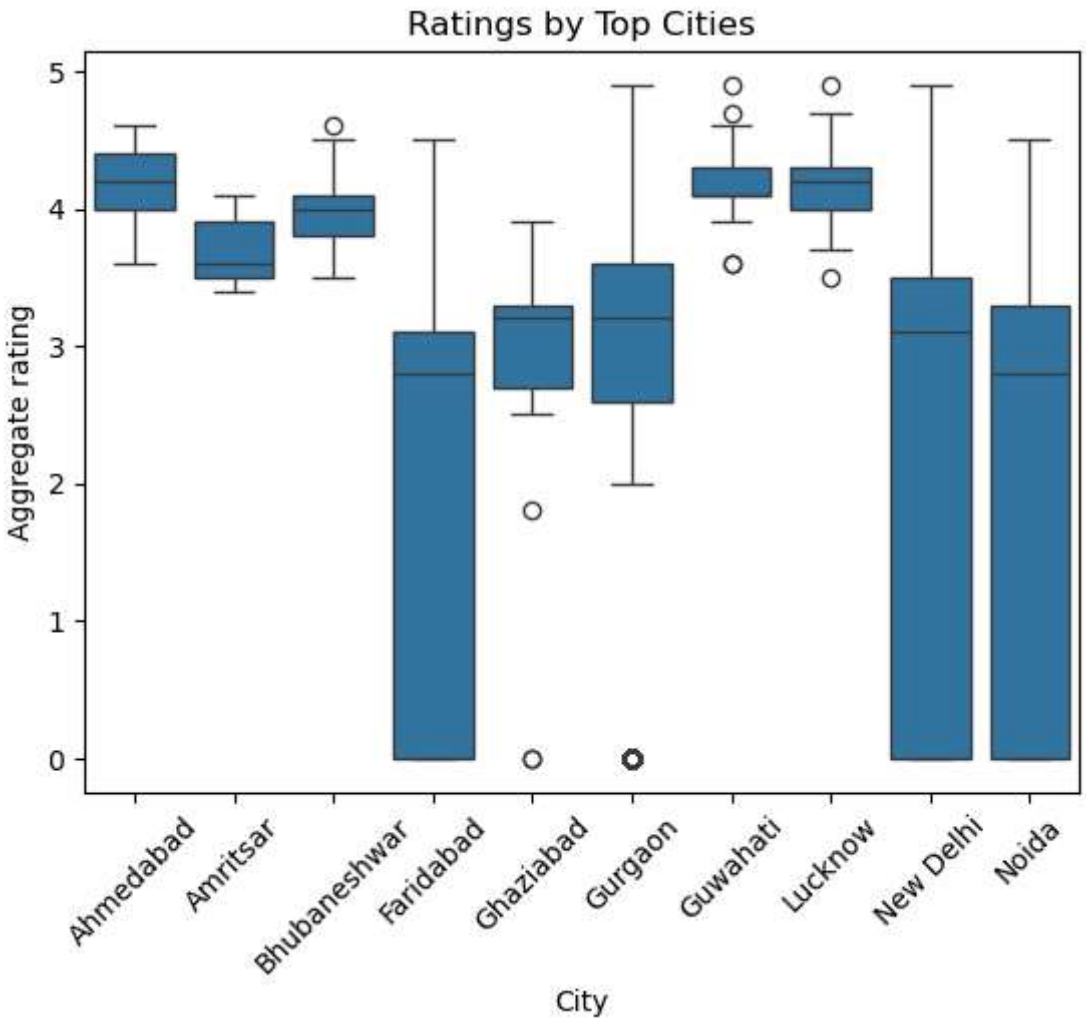
In [9]:
```
plt.figure(figsize=(8,5))
sns.histplot(df['Aggregate rating'], bins=10, kde=True)
plt.title("Distribution of Restaurant Ratings")
plt.xlabel("Rating")
plt.ylabel("Count")
plt.show()
```



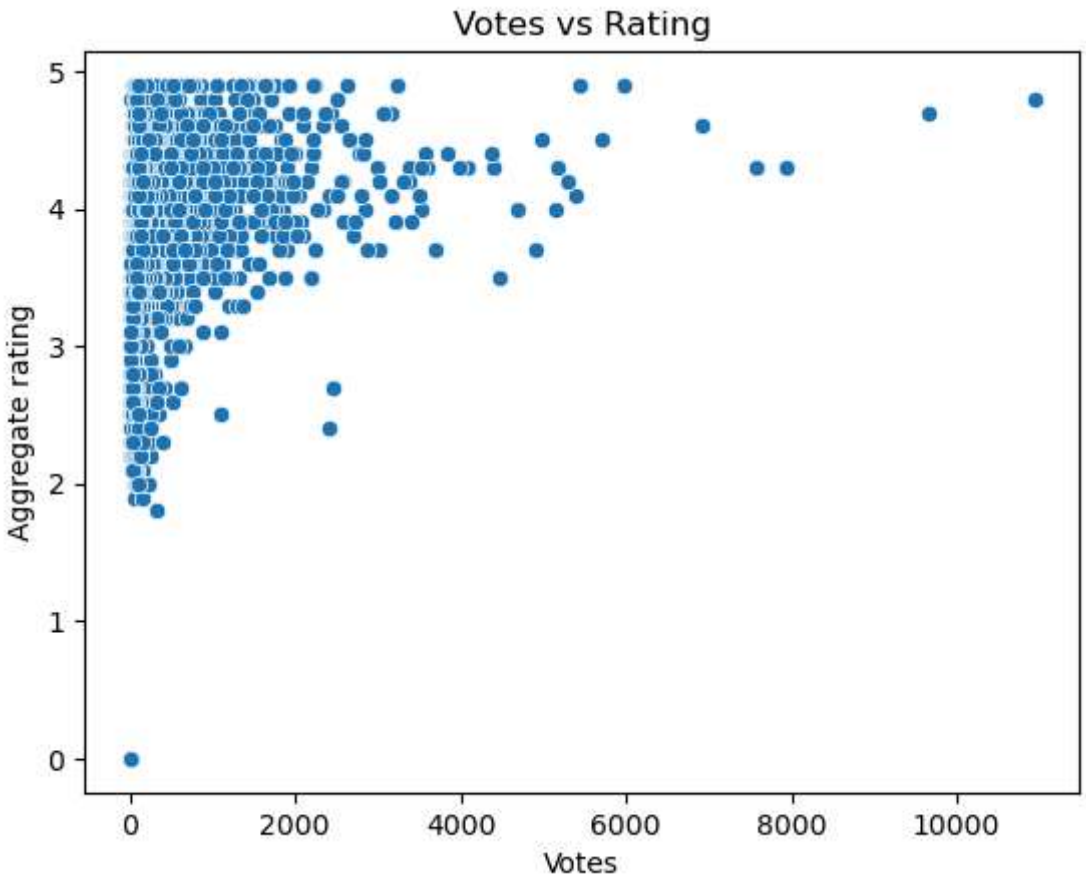
In [10]:
```
df['Cuisines'] = df['Cuisines'].str.split(', ')
df_exploded = df.explode('Cuisines')

top_cuisines = df_exploded['Cuisines'].value_counts().head(10).index

sns.boxplot(x='Cuisines', y='Aggregate rating',
            data=df_exploded[df_exploded['Cuisines'].isin(top_cuisines)])
plt.xticks(rotation=45)
plt.title("Ratings by Top Cuisines")
plt.show()
```



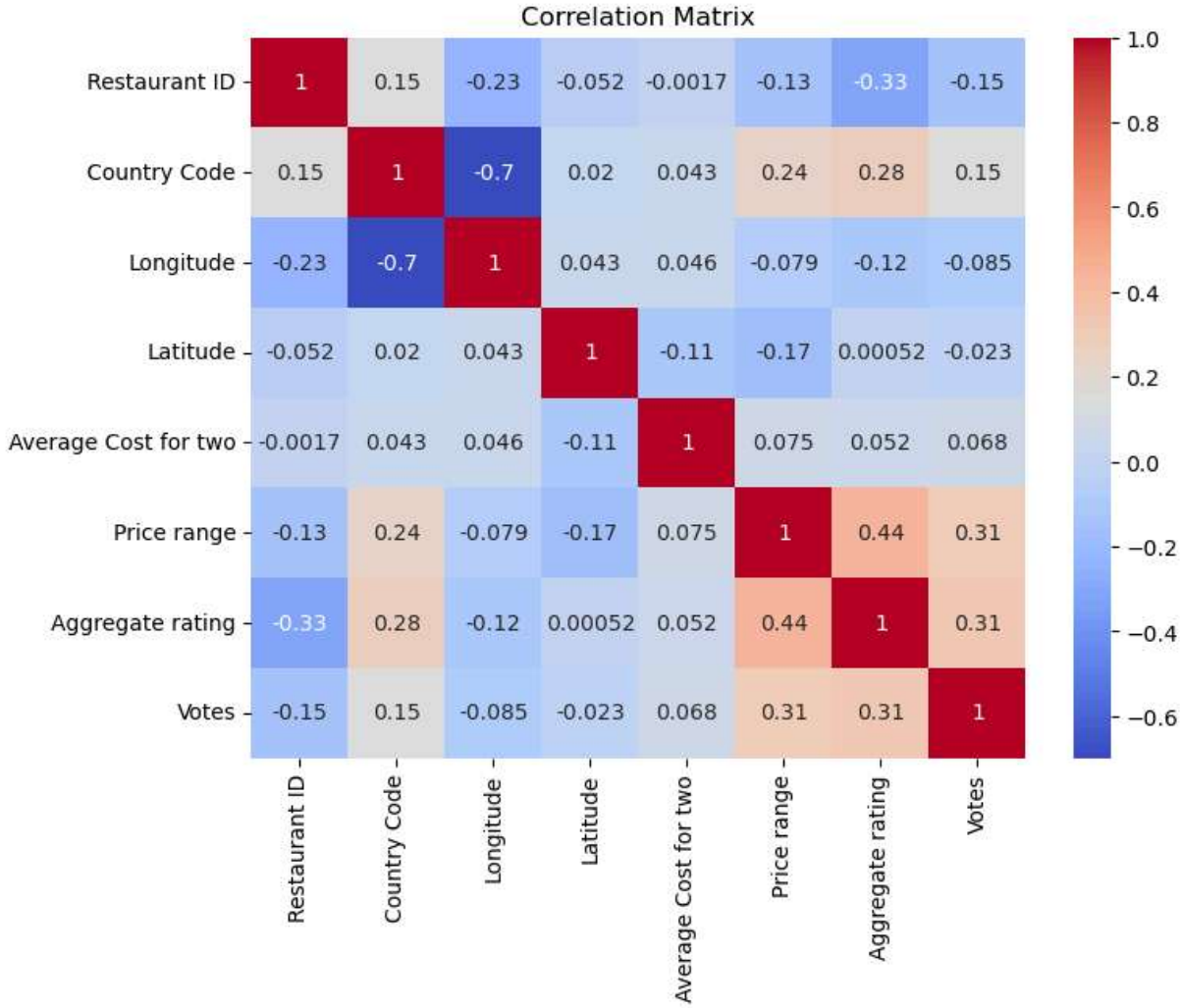
In [11]:
```
top_cities = df['City'].value_counts().head(10).index

sns.boxplot(x='City', y='Aggregate rating',
            data=df[df['City'].isin(top_cities)])
plt.xticks(rotation=45)
plt.title("Ratings by Top Cities")
plt.show()
```



In [12]:
```
sns.scatterplot(x='Votes', y='Aggregate rating', data=df)
plt.title("Votes vs Rating")
plt.show()
```



In [13]:
```
plt.figure(figsize=(8,6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.title("Correlation Matrix")
plt.show()
```


Correlation Matrix
Restaurant ID
Country Code
Longitude
Latitude
Average Cost for two
Price range
Aggregate rating
Votes
1 0.15 -0.23 -0.052 -0.0017 -0.13 -0.33 -0.15
0.15 1 -0.7 0.02 0.043 0.24 0.28 0.15
-0.23 -0.7 1 0.043 0.046 -0.079 -0.12 -0.085
-0.052 0.02 0.043 1 -0.11 -0.17 0.00052 -0.023
-0.0017 0.043 0.046 -0.11 1 0.075 0.052 0.068
-0.13 0.24 -0.079 -0.17 0.075 1 0.44 0.31
-0.33 0.28 -0.12 0.00052 0.052 0.44 1 0.31
-0.15 0.15 -0.085 -0.023 0.068 0.31 0.31 1
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6

In [ ]: